

DEVELOPMENT OF LINE OF THERAPY RULES IN MULTIPLE MYELOMA: OPTUM CLAIMS

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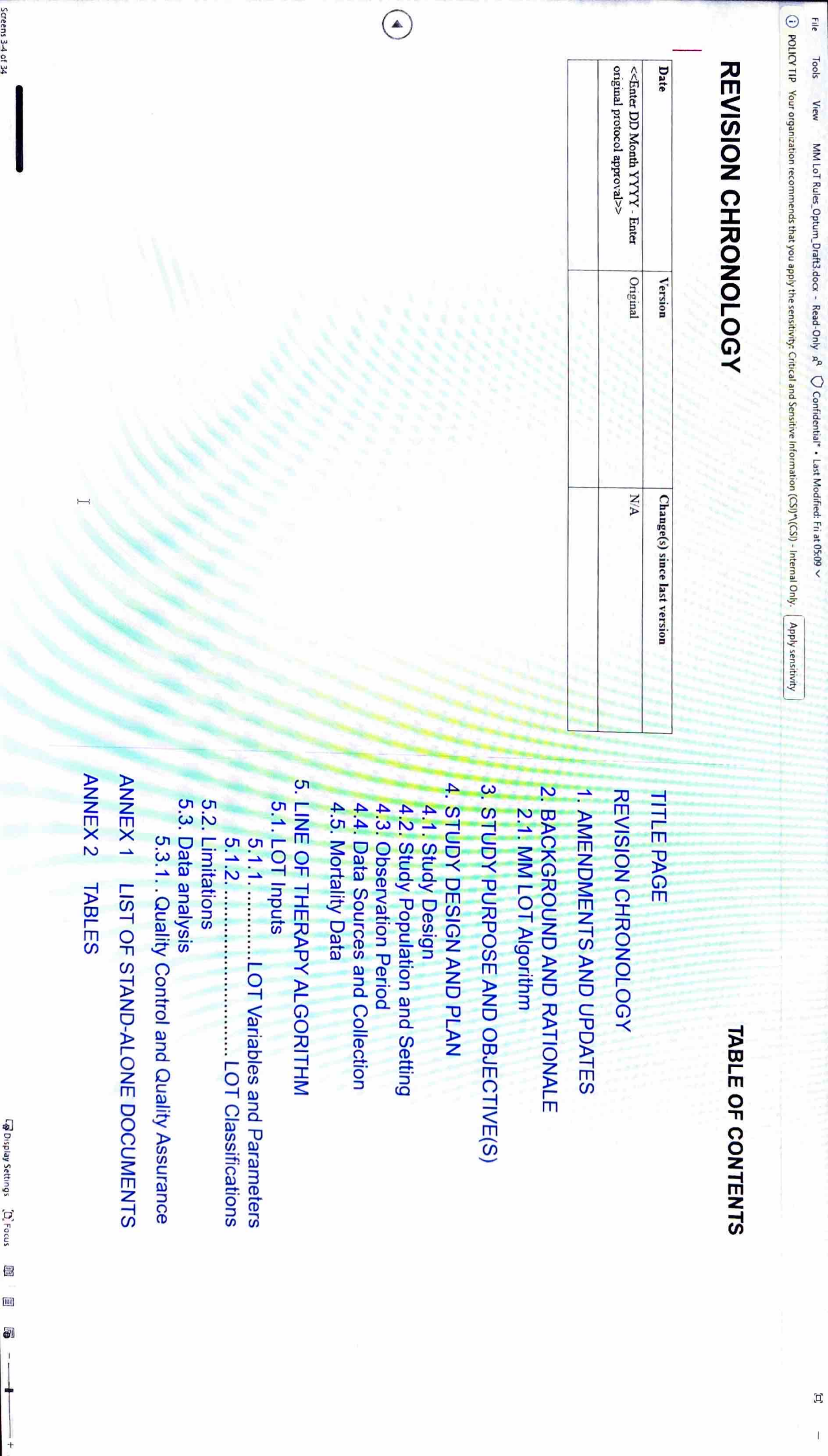
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sequencing, patient comorbidities, and eligibility for stem-cell transplantation (SCT).ⁱⁱ For transplant- ineligible patients, the preferred first-line treatment regimens are daratumumab-lenalidomide-dexamethasone-based regimens; other regimens are recommended or useful in certain circumstances.^{5,6} With an enhanced understanding of the pathophysiology and disease heterogeneity, and the advent of novel treatments, there have been improvements in overall survival of MM patients.ⁱⁱⁱ

Despite the positive trends in disease survival, the natural history of MM results in relapse after initial therapy.^{iv} The choice of treatment from various available regimens to treat relapsed and refractory MM (RRMM) is based on numerous factors. There are no firm treatment guidelines as clinical decisions are based on potential toxicities, patient and physician preferences, prior treatment, cytogenetic risk, comorbidities, myeloma markers and financial considerations.^{v,vi,vii} Treatment options for RRMM patients include re-administration of initial agents for those who are naïve or sensitive to the novel drug combinations and were not exposed to them in the previous six months. Patients who are refractory to the novel drugs should be considered for regimens that do not contain the drugs that they progressed on. Other considerations for RRMM include newer agents in the same class (i.e. carfilzomib and pomalidomide), novel mechanisms of action drugs (XPO inhibitors, BCL-2 inhibitors, cereblon E3 ligase modulators, and T-cell engaging therapies such as bispecific antibodies or CAR-T cell therapy), combination therapies and autologous/allogeneic stem-cell transplants (SCTs) (NB: allogeneic SCTs are rarely done).^{viii} The complexity of the MM treatment landscape, due to multiple treatment phases, gaps between regimens, and combination therapies of three or more drugs, has made it more difficult to identify the real-world clinical pathways of the patients.^{ix,x}

Identifying the prior lines of therapies (LOT) can support selection of treatment regimens for relapsed MM patients, determine eligibility for enrolment into clinical trials, support adherence, and enable accurate health technology assessment for approvals (e.g., FDA).^{xi,xii} As additional novel therapies for MM become available, particularly in the second- and third lines, there is a need to understand various lines of therapy for RRMM patients. The goal of this project is to develop a set of rules to identify lines of therapy for MM patients using administrative claims data. LOT rules will be validated with clinical adjudication and by comparing identified treatment sequences to established external benchmarks and published LOT definitions from registries and pivotal trials.

2.1. MM LOT Algorithm

This document describes an updated version of the MM LOT algorithm rules that GSK and Optum worked to develop. Optum implemented previous versions of the LOT algorithm, for which the output of the first iteration of the MM LOT algorithm was analysed with EHR data and presented to GSK's external experts (EBs). EB feedback led to revisions in the first version of the LOT algorithm, principally, but not limited to, exclusion of corticosteroids by themselves as an MM treatment.

Optum Market Clarity data was used for the original LOT build by Optum, with the future intent of using the linked EHR data to validate the LOT algorithm in claims. It was however found that difference in data granularity and temporality of the claims versus the EHR data made validation not plausible. Feasibility results, based on Market Clarity data, are presented in Appendix A. The identification period in the feasibility was shorter than the identification period

specified in this document because more recent data are available since the feasibility was generated. A table summarizing the demographic profile of patients in Market Clarity, provided by Optum, is presented in Appendix B.

3. STUDY PURPOSE AND OBJECTIVE(S)

The purpose of this project is to develop and validate a LoT algorithm for MM to be used for administrative claims data. The objective of this project is to:

- 1) Develop a LoT algorithm to identify up to five LoTs for MM using administrative claims data.

4. STUDY DESIGN AND PLAN

4.1. Study Design

This retrospective study of newly diagnosed multiple myeloma health patients uses Optum's Clinformatics Data Mart (CDM) for the period of 1 July 2015 to 30 June 2025 (study period). These are the most recent available Optum CDM data at the time of study initiation. Patients with newly diagnosed MM are identified from 1 January 2016 to the end of the study period, 30 June 2025 (identification period). The date of the first claim with a diagnosis code for MM during the identification period is the index date. The baseline is the period of 6 months before the index date. The variable-length follow-up begins on the index date and ends at the earliest of death, health plan disenrollment or the end of the study period.

Starting the study period in 2015 and ending in 2025 will provide nearly ten years of data for patients with index dates earlier in the identification period, which

is a sufficiently long study period to capture patients who are newly diagnosed and develop relapsed/refractory MM.

4.2. Study Population and Setting

The study sample comprises patients with newly diagnosed multiple myeloma in Optum's CDM. The observation period is from 1 July 2015 to 30 June 2025. The identification period for patients with newly diagnosed multiple myeloma is 1 January 2016 to 30 June 2025.

Eligibility Criteria

Each inclusion and exclusion criterion listed below is flexible and may be applied at the study team's discretion to refine the final study population. The sole mandatory requirement for cohort membership is that **patients must have at least one inpatient medical claim or 2 outpatient medical claims for MM in any position on the claim during the study period** (see inclusion criteria #1 for full details).

Inclusion Criteria

Patients are eligible for the project sample if they meet the five inclusion criteria listed below.

- 1. **Newly diagnosed MM:** At least one inpatient medical claim with a diagnosis code for MM (any ICD-9-CM=203.0x or ICD-10-CM code=C90.0x) or ≥ 2 medical claims for MM in any position on the claim,

on separate days within 90 days, during the identification period. (Patients with 2 claims 30 or 60 days apart will also be flagged for consideration). The index date is the date of the first medical claim for MM. Diagnosis codes for MM are in the programming specification document, tab "MM dx code".

As data in Optum is collected primarily for health insurance and not research purposes, all MM codes are to be considered and the code list will not be limited to those specifically indicated for newly diagnosed MM.

2. **Adults:** ≥ 18 years in index calendar year.

3. **Baseline continuous enrollment (CE) of six months:** Six months of CE with medical and pharmacy benefits before the index date. The baseline period is fixed and is the six months before the index date. Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled.

The purpose of the baseline is to create a period during which patients who were previously diagnosed with MM are identified; if treated, these patients will be excluded at a later stage (see Exclusion Criteria below).

4. **Follow-up CE of at least 1 day:** ≥ 1 day of CE starting on the index date until death or study period end. Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled.

- Patients with ≥ 3 months of CE, without gaps in their enrollment, measured from index date will be flagged for consideration for sensitivity or secondary analyses. The follow up period ends on the earliest of health plan disenrollment or the study period end for patients who do not have an assigned date of death < 3 months after the index date.

- For patients with an assigned date of death < 3 months after the index date, these patients are retained and have a limited follow-up period that ends at the assigned date of death.

- As noted in section 7.5, mortality data included month and year of death. The *assigned* date of death for data set construction is set to the 15th of the month.

5. **Evidence of treatment with agents for the purpose of treating MM based on FDA-approval and NCCN guidelines during the follow-up period:** Patients will be required to have evidence of an approved MM oncology therapy in claims during the follow-up period. These will be identified with pharmacy claims (NDC) and medical claims (CPTs and HCPCS) codes.

Receipt of corticosteroids only (e.g., dexamethasone) will not be considered evidence of receiving an approved MM oncology therapy. NDC codes and HCPCS codes for approved MM therapies are in the programming specification document, tabs "MM approved therapy – NDC" and "MM approved therapy – HCPCS", respectively.

Exclusion Criteria

Patients meeting any of the criteria described below are flagged for potential exclusion from the study sample.

1. **Evidence of an MM oncology therapy during the baseline period:** For claims, ≥ 1 medical or pharmacy claim for an FDA-approved MM oncology therapy during the baseline period. The MM oncology therapy

pies are listed in the programming specification document, tab “MMA Codelist”.

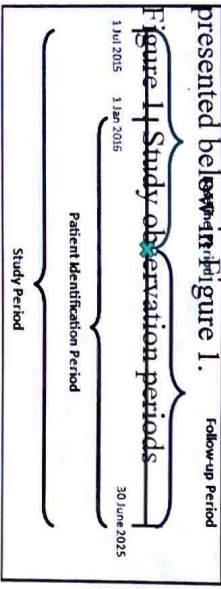
Evidence of another cancer in the baseline period: Patients with either ≥ 1 inpatient or ≥ 2 outpatient ICD-9-CM or ICD-10-CM codes on separate days, within 30 days, for the same primary tumor type and/or metastatic cancer will be excluded. Codes (ICD-9-CM and ICD-10-CM) are in the programming specification document, tab “Other malignancies”.

2. Evidence of pregnancy: ≥ 1 of medical claim with a diagnosis, procedure, or revenue code indicating pregnancy or childbirth during the baseline or follow-up period. Codes for pregnancy are in Appendix B, tab “Pregnancy”.

3. Clinical trial participation: Evidence of clinical trial participation during each of the baseline and follow-up periods will be flagged. Codes for clinical trial participation are in Appendix B, tab “Clinical trial”.

4.3. Observation Period

All patients must have a minimum of 12 months of CE in the health plan: 6 months of baseline and at least 6 months of follow-up. The study periods are presented below in Figure 1.



4.4. Data Sources and Collection

Data for this study were obtained from the Optum Clinformatics® Data Mart (CDM), a HIPAA-compliant, de-identified administrative claims database. The CDM links enrolment and provider information with medical, facility, physician, and pharmacy claims, as well as outpatient laboratory results from major national labs. It includes service dates from January 2007 to the present, covers approximately 15–20 million lives annually across all 50 states, and restricts membership to individuals with both medical and pharmacy benefits.

4.5. Mortality Data

Data on mortality in the Optum Clinformatics Data Mart are captured from various sources to provide the most complete information on death available. As noted in Sections 6.1 and 6.2, assigned date of death was used to as a criterion to end the variable follow-up period.

Year and month of death in the CDM is populated using the following sources and ranking:

1. Centers for Medicare and Medicaid Services (CMS); Mortality information derived from CMS data used in support of Medicare services
2. Social Security Administration Death Master File (DMF)
3. Facility claims where discharge status = expired
4. Member coverage, reason for coverage discontinuation = death
5. Electronic medical records (EMR)
6. Obituary data, externally sourced

5. LINE OF THERAPY ALGORITHM

LOTs are not explicitly captured in claims data and therefore an algorithm based on receipt and timing of therapy is implemented. This section describes the algorithm GSK developed to identify lines of therapy and how the variables in the algorithm are derived. The variables and variable definitions for the LOT algorithm are informed by the clinical literature, input from an Optum expert hematologic oncologist consultant, and the Optum/GSK study team.

The LOT algorithm has been applied to real-world data. As described in [Section 4.1](#), GSK worked with Optum on previous iterations of the MM LOT algorithm. This iteration of the start-up document has been updated by GSK alone to reflect the latest revisions to the LOT algorithm.

For each patient, up to five lines of therapy will be identified for the entire follow-up period. The LOT variables are described immediately below and the

analytic methods for the LOTs, including the LOT validation, are [described in Section 5.2](#).

The MM therapies will be identified from procedure codes (e.g., infusions) in medical settings, including facility outpatient and ambulatory care settings, inpatient hospitalizations, and prescriptions found in claims.

The medications and associated codes used in the LOT algorithm are in the programming specification document, tabs “MM approved therapy – NDC”, “MM approved therapy – HCPCS”, and “Oncology therapies”. All rules described below are based on oncology therapies only (e.g., not corticosteroids).

Note that the same rules and definitions may be described in multiple places. This is to provide the reader with the information to understand and interpret the specific rule being described without needing to refer to other sections of the document.

5.1. LOT Inputs

- **MM therapies:** FDA-approved MM oncology therapy as identified from FDA, NCCN guidelines, and peer-reviewed literature. NDC codes and HCPCS codes for approved MM therapies are in the programming specification document, tabs “MM approved therapy – NDC” and “MM approved therapy – HCPCS”, respectively.
- **Stem cell transplant (SCT):** A LOT-specific binary variable is created to classify whether patients have an SCT during the LOT. SCT is operationalized as ≥ 1 non-diagnostic procedure code for an SCT.

- **Planned Tandem SCT:** Patients will be classified as having a planned tandem SCT if they receive at least two autologous SCTs ≥ 60 to ≤ 180 days apart; otherwise, they will be considered unplanned.²⁰ An autologous SCT followed by an allogeneic SCT will not be considered planned because an allogeneic SCT typically follows failed autologous SCTs. Presence of an unplanned SCT will end the current LOT the day before the unplanned SCT if the LOT has not otherwise ended prior to the date of the unplanned SCT.
- **Allogeneic SCT:** Presence of an allogeneic SCT immediately ends a LOT and starts a new LOT. The new LOT will consist entirely of the allogeneic SCT and no other MM therapies.
- **Maintenance regimen:** A maintenance regimen is a period of 120 days¹ or longer during which only a valid maintenance therapy is available (see list below of valid mono/dual maintenance therapies). Maintenance is not required to be prescribed continuously for the duration of the 120 day period. Furthermore, the maintenance therapy must have been a part of the LOT's initial regimen (for LOT1 the initial regimen includes all MM therapies identified during the first 60 days of LOT1, for LOT2-5 the initial regimen is defined as all MM therapies identified during the first 30 days of the LOT). A maintenance regimen may occur in one of two situations during a LOT:
 1. The LOT's initial regimen transitions into a maintenance regimen, such that non-maintenance medications present during the initial regimen are discontinued leaving only maintenance medications. For example, a patient has an initial regimen of cyclophosphamide and daratumumab, and then discontinues cyclophosphamide leaving only daratumumab,

¹ If an otherwise valid maintenance period that came after an SCT did not reach 120 days due to follow-up ending, then that period is still assigned as maintenance.

which is a valid mono maintenance per the list below.

2. Following an autologous SCT (with 180 days of the HSCT or in the case of a tandem autologous HSCT within 180 days of the second HSCT). The expectation after an HSCT is for a maintenance period and thus instead of the usual 120 days, we check for a valid 30-day (minimum) maintenance period. Any valid maintenance regimens (see list below) present after a single autologous SCT or after the second SCT of a tandem autologous SCT will be considered. These maintenance drugs must have been part of the LOT's initial regimen.

- a. If a full 30-day period is unable to be observed due to the end of follow-up, we will still count whatever days are available as a maintenance period to avoid a new LOT starting immediately before the end of follow-up.

The following will be considered maintenance therapies, per NCCN Guidelines²⁶.

- Lenalidomide monotherapy (NCCN Preferred Regimen for both transplant and non-transplant candidates)
- Bortezomib monotherapy (NCCN Other Recommended Regimen for both transplant and non-transplant candidates)
- Daratumumab monotherapy (NCCN Other Recommended Regimen for transplant candidates)
- Ixazomib monotherapy (NCCN Other Recommended Regimen for both transplant and non-transplant candidates)

- **Thalidomide monotherapy (NCCN Other Recommended Regimen for both transplant and non-transplant candidates)**
- **Bortezomib/lenalidomide dual therapy (NCCN Useful in Certain Circumstances, for non-transplant candidates)**
- **Carfilizomib/lenalidomide dual therapy (NCCN Useful in Certain Circumstances for transplant candidates)**
- **Daratumumab/lenalidomide dual therapy (NCCN Useful in Certain Circumstances for transplant candidates)**

Analysis of real-world data consistently shows that patients may receive regimens that are inconsistent with NCCN guidelines. For example, although daratumumab is recommended for transplant candidates only, it is possible that patients who do not undergo an SCT will receive this medication. Therefore, any of the maintenance medications listed above will be considered maintenance medications for all patients, regardless of transplant status.

Once a maintenance regimen has begun (e.g., observing a valid maintenance medication(s) by itself for at least a 120 day period), it continues until discontinuation (defined below), presence of an SCT, right censoring (i.e. end of follow-up), or the addition of any non-maintenance MM therapy automatically causes the maintenance regimen to end.

5.1.1. LOT Variables and Parameters

- **LOT start date:** LOT 1 begins on the date of the first fill for an MM therapy following a patient's index date. Subsequent LOTs will begin on the earlier of a SCT (allogeneic or an unplanned autologous SCT), CAR-T or the date of the first fill for an MM therapy following the end of the previous LOT.

MM therapies are medications used to treat MM that are FDA-approved, based on NCCN guidelines, and/or reported in the observational peer-review literature.

- **LOT regimen:** A LOT regimen includes all MM therapies received within 60 (30) days^{xiii} on and following the start of LOT 1 (LOT2-5) (see LOT start date above). For LOT 1, the first therapy received is considered the induction therapy.

- The following therapy is excluded from lines of therapy: mesna (treatment of MM to protect the patient's bladder against the toxic side-effects of certain anticancer drugs).

- **LOT end date:** The LOT will continue until the earliest of any of the following:

1. **Substitution** of a new MM treatment after the initial agent(s). For the following permissible substitutions, the line will be named according to the drug given for the longest duration within the line:

- Substitution of rituximab for rituximab/hyaluronidase and vice versa does not advance the LOT
- Substitution of daratumumab for daratumumab/hyaluronidase and vice versa does not advance the LOT
- Substitution of the reference product for a biosimilar or vice versa does not advance the LOT (e.g, trastuzumab and trastuzumab-anns)
- Substitution of bortezomib induction therapy for ixazomib as a maintenance therapy

2. Discontinuation of all agents in the regimen: Discontinuation is defined as the run-out date of all MM treatments in the LOT. Run-out date is the last date that a medication for a specific prescription is considered to be at hand / available.

- The end date is the run-out date before the gap. The run-out date is calculated using a patient's combined infused/injected drugs as well as oral drugs. Overlapping pharmacy benefit drug prescriptions "pushed out" the run-out date. See the LOT1 Specifications Excel document tab "DS_2 PROCESSING" for more details.

- The days supply on medical claims is always missing and therefore a 28 days supply was assumed. When pharmacy claims were missing days supply, or when days supply values were anomalous, the claims were excluded.

3. Unplanned SCTs: If an autologous SCTs occurs > 180 days following a previous autologous SCT in the LOT, then the LOT will end the day before the latter SCT. Any allogeneic SCTs are considered a new LOT and the current LOT will end the day before an allogeneic SCT occurs. Single autologous SCTs and tandem SCTs are considered a continuation of the line of therapy.

4. SCTs not followed by maintenance: If an SCT is not followed by a maintenance regimen within 180 days, then the last day of the LOT is the date of the SCT.

5. Death

6. Health plan disenrollment

7. End of the study period

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8. End of maintenance regimen: The LOT will end after a maintenance regimen ends due to:

- The introduction of any MM therapies including therapies that were part of the original regimen does not advance the LOT (e.g., a LOT has an initial regimen of Bortezomib+Lenalidomide and later the patient has a Bortezomib mono maintenance regimen; if Lenalidomide restarts at some point then the maintenance regimen will end as we observed 120+ days of only Bortezomib and the reintroduction of Lenalidomide could indicate progression).

- Unplanned SCTs (defined above), Death, Health plan disenrollment, end of study period.

Events that will **not** end the LOT are:

- **Single autologous SCT or tandem stem-cell transplants (SCT) $\leq$ 180 days apart:** SCTs are usually preceded by induction therapy and followed by consolidation and maintenance therapy. Tandem SCTs are two SCTs $\geq$ 60 to $\leq$ 180 days apart. These are considered planned and a continuation of the line of therapy. Codes for SCTs are in the programming specification document, tab "SCT".

SCTs must be within 60 days after the end of the induction or maintenance regimen to be considered a continuation of the LOT. This 60-day rule was recommended by Optum's KOL² and informed by Charalampous et al.'s findings that the median time from end of induction to transplant was 33 days (interquartile range=27-42).^{xiv}

² Optum's KOL is Dr Allan Stawis.

- **The start of a maintenance regimen:** Receipt of a valid maintenance regimen (see 4.4.1. for definition)

5.1.2. LOT Classifications

- **First-line (L1; LOT1):** The first line of therapy received by patients with newly diagnosed MM. This includes induction therapy and may also include maintenance and SCTs.

Second-line (L2) and later LOTs: Indicated by initiation of new therapy or an SCT after the end of the previous LOT (1L +/- maintenance or 1L SCT + maintenance). The LOT rules described above apply to LOT2 and later LOTs with the caveat that for LOT2 and later LOTs the regimen identification window (to determine which MM treatments are present in the LOT) is 30 days instead of 60 days.

- **CAR T:** CAR T cellular therapy infusions should be classified as their own LOT.
 - Therapies (including supportive agents, e.g., corticosteroids) given within 30 days of CAR-T cellular therapy are consolidated as part of the CAR T LOT.

5.2. Limitations

- LOTs are not explicitly captured in claims data and the algorithms used to identify lines of therapy are based on multiple clinically valid assumptions.

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- Patients with evidence of MM during the baseline will be excluded from the study sample to ensure that the patients ascertained during the identification period are newly diagnosed with MM. However, because ICD-9-CM and ICD-10-CM codes do not exist to differentiate smoldering from active MM, this will result in the exclusion of patients with smoldering MM who progress to active MM. Therefore, the LOT algorithm developed may not be generalizable to patients who have smoldering MM who progress to active MM.
- Medications with blocked information (e.g., during the time that health insurers are identifying payment and authorization processes) may be missing from Optum's CDM.
- This study's results will be limited by the detail and quality of the structured fields in the databases.

5.3. Data analysis

Overview Analyses will be descriptive and focused on (1) development and performance assessment of the claims-based LOT algorithm, (2) describe real-world treatment pathways/sequencing, and (3) sensitivity and subgroup analyses.

1. The LOT algorithm will be implemented to identify up to five sequential LOTs per patient using the rules described in Section 6.1 and the programming specification document.

- For each LOT, the analysis datasets will include LOT start and end dates, agents/regimens present, and indicators for receipt of CAR-T, type of SCT, and maintenance therapy, as appropriate.

- The distribution of number of LOTs per patient and time between LOTs (mean, standard deviation, median, IQR, min, max) will be tabulated. The most frequent regimens observed in LOT1, LOT2 and subsequent LOTs will be reported with counts and percentages.

2. Descriptive analyses of treatment pathways and outcomes

- Treatment pathways will be described using sequence summaries and visualizations (e.g., Sankey plots) showing transitions across regimen classes and LOTs.
- Utilization of key interventions (SCT, CAR-T, enrollment codes for clinical trials where identifiable) and use of maintenance therapy will be reported as counts (n, %) and descriptively (median [IQR], mean [SD]).

3. Subgroup and sensitivity analyses

- Sensitivity analyses will assess robustness of LOT rules by varying key algorithm parameters (e.g., required amount of continuous enrollment in the follow up period, inclusion/exclusion of corticosteroid-only claims, look-back windows for prior exposure) and by applying alternative cohort definitions (e.g., stricter continuous enrollment criteria, restricting the patient population by the 'Additional Eligibility Criteria' listed in section 5.2). Effects on primary outputs (number of LOTs, timing, regimen classification) and on validation concordance metrics will be reported.

5.3.1. Quality Control and Quality Assurance

Validation of the LOT algorithm will include clinical adjudication and comparing the finalized LOT to external benchmarks. These benchmarks include

claims-derived treatment sequences, regimen frequencies and LOT distributions. Outputs will be compared to published LOT definitions, registry data and pivotal trial reporting. Descriptive comparisons and standardized differences to be reviewed by the study team.

ANNEX 1 LIST OF STAND-ALONE DOCUMENTS

The table below should be used to identify standalone documents. Indicate <None> if there is none.

No.	Document Reference No	Date	Title
1.	<<No>>	<<Enter DD Month YYYY>>	<<Text>>
2.	<<No>>	<<Enter DD Month YYYY>>	<<Text>>
3.	<<No>>	<<Enter DD Month YYYY>>	<<Text>>

ANNEX 2 TABLES

Appendix A. Original Feasibility in Market Clarity

These counts were generated by Optum during the proposal process and are for information purpose only.

Criteria	Number of patients
Inclusion	
Patients with at least 1 non-diagnostic claim for multiple myeloma (ICD-9-CM=203.00 or ICD-10-CM=C90.00) during 1 January 2016 to 31 December 2021 (the identification period). The first date is the index date	95,790
Continuous enrollment: 6 months before index date (Baseline period)	35,930
Continuous enrollment: ≥3 months after index date, starting on index date (Follow-up period)	32,945
Age ≥18 years as of the index date	32,825
At least 1 claim for systemic MM cancer therapy on or after the index date. The date of the first claim is the first line of therapy start date	18,114
No evidence of systemic cancer therapy in baseline period	11,958
No evidence of MM in baseline period	7,971
No evidence of R/R MM (≥2 claims on different dates of service), in baseline period	7,864
No evidence of other cancers in baseline period	5,971

Counts among 5,571 patient population	
Patients with information on prescriptions written/administered in EHR	2,467
Patients with at least 1 non-missing prescriptions written or medications administered occurring on or after LOT1 start date	1,853
Patients with at least 5 non-missing prescriptions written or medications administered on different dates of service occurring on or after LOT1 start date	
Patients with physician description of treatment rationale in EHR	
Patients with at least 1 non-missing discontinuation reason (e.g., "lack of efficacy," "side effect," "alternative therapy prescribed") occurring on or after LOT1 start date	1,685
Patients with at least 2 non-missing discontinuation reasons (e.g., "lack of efficacy," "side effect," "alternative therapy prescribed") on different dates of service occurring on or after LOT1 start date	1,506
Patients with physician description of drug action in EHR	
Patients with at least 1 non-missing drug action (e.g., "start," "stop," "switch") occurring on or after LOT1 start date	2,003
Patients with at least 2 non-missing drug actions (e.g., "start," "stop," "switch") on difference dates of service occurring on or after LOT1 start date	1,730

Appendix B. Characteristics of patients in Market Clarity
Demographics of patients in Market Clarity are presented below in Table 1.
These counts were generated by Optum during the proposal process and are for information purpose only.

Table 1. Characteristics of Patients in Market Clarity and Population in US Census

Characteristic	Market Clarity ⁽¹⁾ (Jan 2007 – Jun 2022)		US Census Population ⁽²⁾ (2021)	
	Number of patients (million)	%	Number of people (million)	%
Total	76.8	100%	328.1	100%
Insurance Type				
Commercial	48.8	64%	216.4	66%
Medicaid	13.5	18%	61.9	19%
Medicare	13.4	17%	60.2	18%
Medicare Advantage + Part D	n/a	-	n/a	-
Other Payer Type	4.8	6%	3.2	1%
Uninsured	15.0	20%	27.2	8%
Unknown	10.1	13%	n/a	-
Census Region				
Midwest	30.4	40%	67.9	21%
Northeast	13.9	18%	56.1	17%
South	17.2	22%	126.1	38%
West	11.1	14%	77.9	24%
Unknown ⁽³⁾	4.1	5%	n/a	n/a
Age Cohort (years) ⁽⁴⁾				
0-18	10.2	13%	77.0	23%
19-25	5.6	7%	29.0	9%
26-34	9.3	12%	40.3	12%
35-54	20.2	26%	125.5	38%
55-64	10.6	14%		
65+	20.9	27%	56.2	17%
Unknown ⁽⁴⁾	0.0	0%	n/a	n/a
Gender				
Female	41.0	53%	166.2	51%
Male	35.8	47%	160.0	49%
Unknown	0.0	0%	n/a	n/a

Characteristic	Market Clarity ⁽¹⁾ (Jan 2007 – Jun 2022)		US Census Population ⁽²⁾ (2021)	
Race/Ethnicity				
African American	7.4	10%	44.0	13%
Asian	1.8	2%	20.7	6%
Caucasian	45.8	60%	194.2	59%
Hispanic	4.5	6%	62.5	19%
Other/Unknown	17.3	22%	6.7	2%

⁽¹⁾ In The Optum[®] re-identified Market Clarity dataset

⁽²⁾ US Census Health Insurance Coverage in the United States: 2021 (census.gov)

⁽³⁾ Age is calculated as 2022 minus year of birth.

⁽⁴⁾ Unknown reflects instances when data fell outside of the defined range.

⁽¹⁾ The Optum® de-identified Market Clarity dataset
⁽²⁾ Census: Health Insurance Coverage in the United States 2022 Health Insurance Coverage in the United States.
⁽³⁾ Age is calculated as 2022 minus year of birth.
⁽⁴⁾ Unknown reflects instances when data fell outside of the defined range.

Appendix C. Pressure-testing Patient Identification Criteria

Additional steps, summarized below, were implemented by Optum's team using the Optum Market Clarity data to confirm patients' MM diagnosis and to identify and exclude those with relapsed MM. These steps were supplemental to the inclusion and exclusion criteria and, ultimately, did not change the original sample size of N = 5,372 patients.

- **EHR flags for MM diagnosis date:** Patients meeting the following three criteria in the EHR diagnosis table were be flagged: 1) diagnosis of ICD-9-CM=203.00 or ICD-10-CM code=C90.00; 2) diagnosis date +/- 30 days of the index date; 3) diagnosis status of "possible diagnosis of", "diagnosis of", "not recorded".
1. This step was used to confirm the claims-based diagnosis of newly diagnosed MM.
- **Claims flags for longer baseline periods:** Patients with each of 9, 12, and 18 months of baseline continuous enrollment were flagged. These longer

baseline periods were considered to maximize identification of patients with MM patients before the identification period and determine the resulting sample size after these patients were excluded. GSK and Optum decided on a 6-month baseline period.

- **Claims flag for minimum 3-month follow-up period:** GSK and Optum explored a minimum 3-month follow-up period for all patients and decided on the minimum 6-month variable follow-up except for patients who died less than 6 months after the index date.

- **EHR flags for approved MM therapies:** Patients with ≥ 1 EHR record for approved MM therapies from the EHR prescriptions written, medications administered, and procedure tables were flagged. Approved MM therapies in the EHR tables were measured with the same NDC, CPT, and HCPCS codes applied to the claims data. These flags were used to identify the potential number of patients available for the EHR validation of the claims-based LOT.

- **Flags for baseline MM diagnoses:** Patients with ≥ 1 non-diagnostic medical claim for MM during 9-month, 12-month, and 18-month baseline periods, patients with ≥ 1 EHR diagnosis table record with a diagnosis of MM during 6-month, 9-month, 12-month, and 18-month baseline periods, and patients with ≥ 1 EHR diagnosis table record with diagnosis_status = "possible diagnosis" of MM during 6-month, 9-month, 12-month, and 18-month baseline periods were identified.

- Exclusion of patients with baseline MM may have excluded those with smoldering MM, which can be a precursor to active MM: a specific diagnosis code does not exist to differentiate smoldering from active MM. The implication of this exclusion is described in the limitations section.

- **Flags for baseline R/R MM:** Patients with ≥ 1 non-diagnostic medical claim for R/R MM during 9-month, 12-month, and 18-month baseline periods, patients with ≥ 1 EHR diagnosis table record with a diagnosis of R/R MM during 6-month, 9-month, 12-month, and 18-month baseline periods, and patients with ≥ 1 EHR diagnosis table record with diagnosis_status = "history of" of R/R MM during 6-month, 9-month, 12-month, and 18-month baseline periods were identified.

- **EHR flags for other baseline cancers:** Patients with ≥ 2 EHR diagnosis table records on separate dates and within 30 days of each other for the same primary tumor type, for metastatic cancer, and for a primary cancer and metastatic cancer were identified.

- **Flags for claims and EHR records with diagnoses for other cancers during follow-up:** Patients with ≥ 2 medical claims or EHR diagnosis table records on separate dates and within 30 days of each other for the same primary tumor type, for metastatic cancer, and for a primary cancer and metastatic cancer were identified. These patients were identified because the LOT algorithm was developed for patients with MM only; those with other cancers may have received different treatments. GSK and Optum decided not to exclude patients with other cancers in the follow-up period.

6. REFERENCES

1. Rajkumar SV. Multiple myeloma: 2020 update on diagnosis, risk-stratification and management. *Ann J Hematol*. 2020;95(5):548-67.

